Internship Report

On

Customer Shopping Behavior Analysis using Python, SQL & Power BI

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: CS-2341887
Name of Student: ALI GHAZANFAR

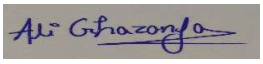
Batch: 2023-27

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CANDIDATE'S DECLARATION

I, ALI GHAZANFAR, do hereby declare that the internship report titled “Customer Shopping Behavior Analysis using Python, SQL & Power BI” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: 

Student Name: ALI GHAZANFAR

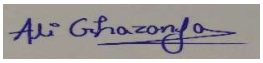
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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AICTE–EduSkills Virtual Internship Program in Data Analytics with Python & Power BI. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to my mentors and the EduSkills Academy team for sharing their truthful and illuminating views on a number of issues related to this work.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me the opportunity to undertake this internship as part of my academic curriculum.

Finally, I express my gratitude to my parents and friends for their constant support and blessings.

Signature: 

Student Name: ALI GHAZANFAR

Roll No.: CS-2341887

INTERNSHIP OFFER LETTER / PROOF OF INTERNSHIP

The AICTE–EduSkills Internship Offer Letter (Ref No. C18Y2026M081546344) confirming enrolment in the 8-week Virtual Internship Program is provided below. The final Internship Completion Certificate will be attached separately upon successful completion of all assessments.

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Ref No: C18Y2026M081546344

Internship Offer Letter

Student Name	: Ali Ghazanfar
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Data Analytics with Python & Power BI
Internship Duration	: August 2026 - October 2026
AICTE Student ID	: STU6997660ac608d1771529738

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Data Analytics Foundations & Python Basics	Course overview, environment setup (Python/Anaconda, SQL, Power BI), core Python concepts (variables, control flow, functions), NumPy, and Pandas fundamentals.
Week 2	Data Cleaning & SQL Fundamentals	Data preprocessing with Pandas (missing values, outliers, transformations), SQL basics including SELECT, WHERE, GROUP BY, aggregations, and joins.
Week 3	Advanced SQL & Power BI Basics	Subqueries, CTEs, window functions, DML operations, introduction to Microsoft Power BI interface, data connection, and basic visualizations.
Week 4	Python–SQL Integration & Data Modeling	Python database connections, executing queries, automation workflows, Power BI data modeling, DAX basics, measures, and time intelligence functions.
Week 5	Advanced Visualization with Python & Power BI	Advanced Power BI visuals (drill-through, bookmarks, filters), performance optimization, Python visualization using Matplotlib and Seaborn for exploratory data analysis.
Week 6	Integration, Publishing & Collaboration	Python scripting integration in Power BI, advanced use cases, publishing dashboards to Power BI Service, sharing, collaboration, and workspace management.
Week 7	Real-World Projects & Workflow Best Practices	Sales performance analysis, customer behavior segmentation, structured workflow practices, and best practices in Python, SQL, and Power BI integration.
Week 8	Capstone Project – Data Preparation & SQL AnalysisCapstone Project – Dashboard & Insights Delivery	E-commerce dataset cleaning using Python, SQL-based extraction and transformation, KPI identification, building dashboards in Power BI, generating insights, optimization, and final presentation.
Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Under the Patronage of AICTE, NATIONAL INTERNSHIP PORTAL

4. Project Description

4.1 Introduction

This report documents the work undertaken during the 8-week AICTE–EduSkills Virtual Internship Program in Data Analytics with Python & Power BI (August 2026 – October 2026), completed under the guidance of the internship mentors as part of the B.Tech CSE curriculum at IILM University, Greater Noida.

The internship followed a structured, week-wise curriculum covering Python programming fundamentals, data cleaning, SQL, and Power BI dashboarding, culminating in a capstone project. As the capstone deliverable, a complete end-to-end data analytics project titled "Customer Shopping Behavior Analysis" was carried out, covering data cleaning and exploration in Python, structured business-question analysis using SQL, and the design of an interactive Power BI dashboard.

The project analyzes a transactional retail dataset of 3,900 customer purchases to uncover insights into customer spending patterns, product preferences, discount behavior, and subscription trends, and translates these insights into actionable business recommendations.

4.2 Organization Profile

The internship was conducted under the AICTE (All India Council for Technical Education) – EduSkills Virtual Internship Program, supported by EduSkills Academy. EduSkills is an ISO 9001:2015 and ISO 20000-1:2018 certified organization recognized as a "Great Place to Work," operating under the patronage of AICTE and the National Internship Portal, with a mission of "Nation Building Through Skills."

EduSkills partners with academic institutions across India to deliver industry-relevant, skill-based virtual internship programs in emerging technology domains such as Data Analytics, Cloud Computing, Artificial Intelligence, and Cybersecurity, enabling students to gain practical, hands-on exposure alongside their formal education.

4.3 Problem Statement

Retail businesses generate large volumes of transactional data, but without structured analysis this data rarely translates into actionable insight. The organization in this

project needed a clear understanding of who its customers are, how they shop, which products and categories perform best, how discounts and subscriptions affect revenue, and which customer segments and shipping preferences drive the most value — information required to guide targeted marketing, inventory, and loyalty strategies.

4.4 Project Objectives

- To clean, prepare, and explore a raw customer shopping transactional dataset using Python.
- To perform structured business-question analysis using SQL on a relational (PostgreSQL) database.
- To identify revenue patterns across gender, age group, subscription status, and shipping type.
- To segment customers into New, Returning, and Loyal categories based on purchase history.
- To identify top-performing and discount-dependent products.
- To design an interactive Power BI dashboard summarizing key business metrics.
- To translate analytical findings into actionable business recommendations.

4.5 Scope of the Project

The project scope covers the complete analytics pipeline for the given dataset: data ingestion and cleaning in Python, feature engineering, database integration with PostgreSQL, SQL-based business analysis, and dashboard development in Power BI. The dataset used contains 3,900 rows and 18 columns spanning customer demographics, purchase details, and shopping behavior. The scope is limited to the dataset provided and does not extend to real-time data pipelines or predictive modeling, though these are noted as possible future extensions.

4.6 Technologies and Tools Used

Category	Tools / Technologies
Programming Language	Python (Pandas, NumPy)
Data Visualization (EDA)	Matplotlib, Seaborn
Database	PostgreSQL

Category	Tools / Technologies
Query Language	SQL (subqueries, CTEs, window functions, joins)
Business Intelligence	Microsoft Power BI
Development Environment	Jupyter Notebook / Anaconda

4.7 System Architecture / Workflow

The project followed a linear data-analytics workflow, illustrated below:

Raw Dataset (CSV) → Data Cleaning & Feature Engineering in Python (Pandas) → Load into PostgreSQL Database → SQL-based Business Analysis → Power BI Dashboard → Business Insights & Recommendations.

Each stage fed into the next: the cleaned and standardized DataFrame produced in Python was loaded into PostgreSQL for structured SQL querying, and the query outputs were then used to build the visual dashboard in Power BI.

4.8 Methodology

Data Loading and Initial Exploration: The dataset (3,900 rows, 18 columns) was imported using pandas. Structural checks were performed using df.info() and summary statistics using .describe().

Missing Data Handling: 37 missing values were identified in the Review Rating column and were imputed using the median rating of the corresponding product category.

Column Standardization: All column names were converted to snake_case for consistency and readability.

Feature Engineering: An age_group column was created by binning customer ages, and a purchase_frequency_days column was derived from purchase-frequency data.

Data Consistency Check: The discount_applied and promo_code_used columns were checked for redundancy; promo_code_used was dropped as duplicate information.

Database Integration: The cleaned DataFrame was loaded into a PostgreSQL database to enable structured SQL analysis.

SQL-Based Business Analysis: Ten business questions were answered using SQL, including revenue by gender, high-spending discount users, top-rated products,

shipping-type comparison, subscriber vs. non-subscriber spend, discount-dependent products, customer segmentation, top products per category, repeat-buyer subscription likelihood, and revenue by age group.

Dashboard Design: An interactive Power BI dashboard was built with slicers for subscription status, gender, category, and shipping type, alongside KPI cards and category-wise / age-group-wise revenue and sales visuals.

4.9 Expected Outcomes / Results

The analysis produced the following key results:

- Male customers generated substantially higher total revenue (₹157,890) than female customers (₹75,191).
- 839 customers were identified as high-spending discount users — spending above the average purchase amount even after applying a discount.
- Gloves, Sandals, Boots, Hat, and Skirt received the highest average review ratings (3.78–3.86).
- Express shipping orders had a marginally higher average purchase amount (₹60.48) than Standard shipping (₹58.46).
- Non-subscribers accounted for the larger share of total revenue (₹170,436 from 2,847 customers) compared to subscribers (₹62,645 from 1,053 customers), though average spend per customer was similar across both groups.
- Hat, Sneakers, Coat, Sweater, and Pants were the most discount-dependent products, with 47–50% of their purchases involving a discount.
- Customer segmentation classified 3,116 customers as “Loyal,” 701 as “Returning,” and 83 as “New.”
- Jewelry, Blouse, and Sandals emerged as the top-selling products in the Accessories, Clothing, and Footwear categories respectively.
- 2,518 non-subscribers versus 958 subscribers were classified as repeat buyers (customers with more than five purchases).
- The Young Adult age group contributed the highest total revenue (₹62,143), followed closely by Middle-aged, Adult, and Senior groups.

Based on these findings, the following business recommendations were proposed: boosting subscription uptake through exclusive member benefits, introducing loyalty programs to convert returning customers into loyal ones, reviewing the discount policy to balance sales growth with profit margins, prominently featuring top-rated and best-selling products in marketing campaigns, and focusing targeted marketing on high-revenue age groups and express-shipping users.

4.10 Certificates of Completion and Communication Proof

The AICTE–EduSkills Internship Offer Letter confirming enrolment in this internship program has been attached in the front matter of this report (Section 3: Internship Offer Letter / Proof of Internship). As stated in the offer letter, this document will be treated as a valid record of internship enrolment only after successful completion of all assessment requirements, following which the final Virtual Internship Completion Certificate will be issued and shall be attached here as supporting proof.

5. Bibliography / References

1. AICTE–EduSkills Virtual Internship Program, Internship Offer Letter, Ref No. C18Y2026M081546344, EduSkills Academy, 2026.

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